AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 20 · STUDENT LESSON

# Data retention and deletion

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

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## Before you start

Recall Sandboxing and resource limits. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

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STUDENT LESSON · 1 OF 18

## Your mission

> Start with the target: Explain the mechanism and terminology of data retention and deletion. Try the recall before opening notes.

### Your learning target

I can explain the **mechanism** and terminology of data retention and deletion.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Sandboxing and resource limits. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from External side effects and recovery. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of data retention and deletion

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Retrieve what you already know

Recall Sandboxing and resource limits. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

### This week's end goal

Explain data retention and deletion using the deciding step in this fictional service.

## Notice the evidence

> Investigate the supplied Data retention and deletion case. Separate observations, given facts and assumptions before choosing an explanation.

Retention and deletion cover all declared stores, including caches and logs. Removing a visible row may leave copies elsewhere.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

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## See how it works

> Follow the mechanism for Data retention and deletion. Connect each labelled input, deciding step and result.

Toy policy retains session content for 7 days and deletes it when age is at least 7. Audit metadata may remain for 30 days without content. Track primary storage, cache and backup policy separately.

**Data retention and deletion — evidence trace**

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Record ages are 6, 7, 9 days. Cache contains copies of all three. | Toy policy retains session content for 7 days and deletes it when age is at least 7. | Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry process. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

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## Compare the cases

> Compare the supplied cases for Data retention and deletion. Identify the detail that changes the conclusion.

### Compare a familiar case with a changed case

Record ages are 6, 7, 9 days. Cache contains copies of all three. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

### Change one thing in the pictured case

#### Change one thing

Change: Primary deletion succeeds but backup deletion fails.

#### Predicted result

**Prediction**: Record incomplete deletion and follow the retry or escalation procedure.

#### Why the result follows

Reason: One required copy remains in the declared scope.

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## Words that unlock the idea

Use the vocabulary of Data retention and deletion to explain a concrete decision, rather than only memorising definitions.

| Word | Meaning |
| --- | --- |
| Retention period | The declared lifetime of stored information |
| Deletion boundary | The age at which removal applies |
| Residual copy | A remaining duplicate in another store |

**Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

## Follow a worked example

Follow the worked trace for Data retention and deletion, predict a step, then compare with its explanation.

Record ages are 6, 7, 9 days. Cache contains copies of all three.

Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry **process**.

### Worked explanation · Data retention and deletion

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Data retention and deletion

Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups.

### The facts we will use

#### Supplied facts

T1 is six days old. T2 is seven days old with one backup. Scope lists primary storage and that backup only.

AI ACADEMY / AI-6/20

# Data retention and deletion

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Check age

Keep T1 for now; T2 reaches the deletion boundary.



## 2. Find copies

T2 is present in two declared locations.



## 3. Verify removal

The scoped deletion passes only after both T2 copies are removed.

**CHANGE ONE THING**

Primary deletion succeeds but backup deletion fails.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Check age | Keep T1 for now; T2 reaches the deletion boundary. |
| 2. Find copies | T2 is present in two declared locations. |
| 3. Verify removal | The scoped deletion passes only after both T2 copies are removed. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

Retention requires an age rule and an inventory. Deleting the visible copy alone does not demonstrate removal of listed backups.

### What this example does not establish

#### Keep the stated limit

The inventory cannot prove deletion from unknown locations; document scope and investigate untracked copies.

### Keep the stated limit

Read the labelled picture: Apply the seven-day boundary to T1 and T2. How many T2 locations need verification, and may deletion be called complete if its backup remains?

### Keep the stated limit

Change one thing: Primary deletion succeeds but backup deletion fails. Predict the result before checking the explanation. What remains the same?

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STUDENT LESSON · 7 OF 18

## Find and repair a mistake

> Find the unsupported step in the Data retention and deletion claim and explain a repair using the supplied evidence.

This fictional retention exercise is not a statement of legal retention requirements.

### Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

### Locate the earliest unsupported step

A report omits this qualification: This fictional retention exercise is not a statement of legal retention requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**State one repair and the checks you would rerun.**

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STUDENT LESSON · 8 OF 18

## Practise together

Practise Data retention and deletion with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

**Explain your prediction to a partner before comparing results.**

### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Data retention and deletion”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
| --- | --- | --- |
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Data retention and deletion. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

> Apply the seven-day boundary to T1 and T2. How many T2 locations need verification, and may deletion be called complete if its backup remains?

**My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

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STUDENT LESSON · 9 OF 18

## Run the investigation

> Investigate Data retention and deletion. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

### Predict, investigate and preserve the evidence

Record ages are 6, 7, 9 days. Cache contains copies of all three. Show the intermediate steps and give the conclusion with its scope.

| Record | Your evidence |
| --- | --- |
| Given input and fixed rule | |
| Prediction before checking | |
| Actual result, or “not run” | |

| Record | Your evidence |
| --- | --- |
| First mismatch and one explanation | |
| Boundary or missing input and expected response | |

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

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STUDENT LESSON · 10 OF 18

## Build your understanding

> Move from recall to tracing, diagnosis and transfer in Data retention and deletion. Explain why each step is justified.

**Explain → apply → challenge**

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: This fictional retention exercise is not a statement of legal retention requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**Explain which evidence supports your conclusion and which claim remains unproved.**

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# Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

Add a labelled data retention and deletion **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of data retention and deletion.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

| Project evidence | Record |
| --- | --- |
| Component and version | |
| Inputs permitted at prediction time | |
| Trace and comparison with earlier work | |
| Failure, limit and human review | |

## Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

## Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My limited claim and a useful next check**

## Study a complete case

> Revisit the complete Data retention and deletion case for supported practice. Repeated examples do not become fresh assessments.

### Complete evidence record

**Data retention and deletion — evidence trace**

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Record ages are 6, 7, 9 days. Cache contains copies of all three. | Toy policy retains session content for 7 days and deletes it when age is at least 7. | Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry process. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

# Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

### Independent case: Changed independent case

Fictional retention policy deletes records at age >=4 days. X is3 days, Y4, Z5. A copy of Y remains in a temporary log after its main record is deleted. Evaluate compliance.

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Attempt this without hints

Retention is5 days, inclusive **deletion boundary**. Ages 4, 5, 6; a log stores complete content. Trace required action.

**My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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## Your lesson route

> Use this map to navigate Data retention and deletion, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups.

| By the end, I can | How I will show it |
|---|---|
| Explain the mechanism and terminology of data retention and deletion | Explain the deciding step, show a trace or test, and state one limit. |
| Apply the stated procedure to a new case with a traceable result | Explain the deciding step, show a trace or test, and state one limit. |
| Identify an assumption or limitation and design a revealing follow-up | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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STUDENT LESSON · 16 OF 18

## Finish and return

> Explain Data retention and deletion without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of data retention and deletion

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

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STUDENT LESSON · 17 OF 18

# Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? The scoped deletion passes only after both T2 copies are removed.

1. Check age

2. Find copies

3. Verify removal

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Verify removal The scoped deletion passes only after both T2 copies are removed. Retention requires an age rule and an inventory.

Deleting the visible copy alone does not demonstrate removal of listed backups.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: Primary deletion succeeds but backup deletion fails. Which conclusion is supported by the supplied facts?

1. Record incomplete deletion and follow the retry or escalation procedure.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Record incomplete deletion and follow the retry or escalation procedure. One required copy remains in the declared scope.

### Repair a missing building block

Review Sandboxing and resource limits (ai-6/19). Goal: Explain the **mechanism** and terminology of sandboxing and resource limits. Short practice: Check A against both resource limits and B against network isolation.

Why must A stop after three seconds even if its **output** remains only 2 MB?. Return to this week and explain what you changed.

Review External side effects and recovery (ai-6/17). Goal: Explain the **mechanism** and terminology of external side effects and recovery. Short practice: Distinguish a missing printer acknowledgement from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?. Return to this week and explain what you changed.

Review Planning and action selection (ai-6/14). Goal: Explain the **mechanism** and terminology of planning and action selection. Short practice: Choose a permitted plan when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?. Return to this week and explain what you changed.

### Keep your completed work

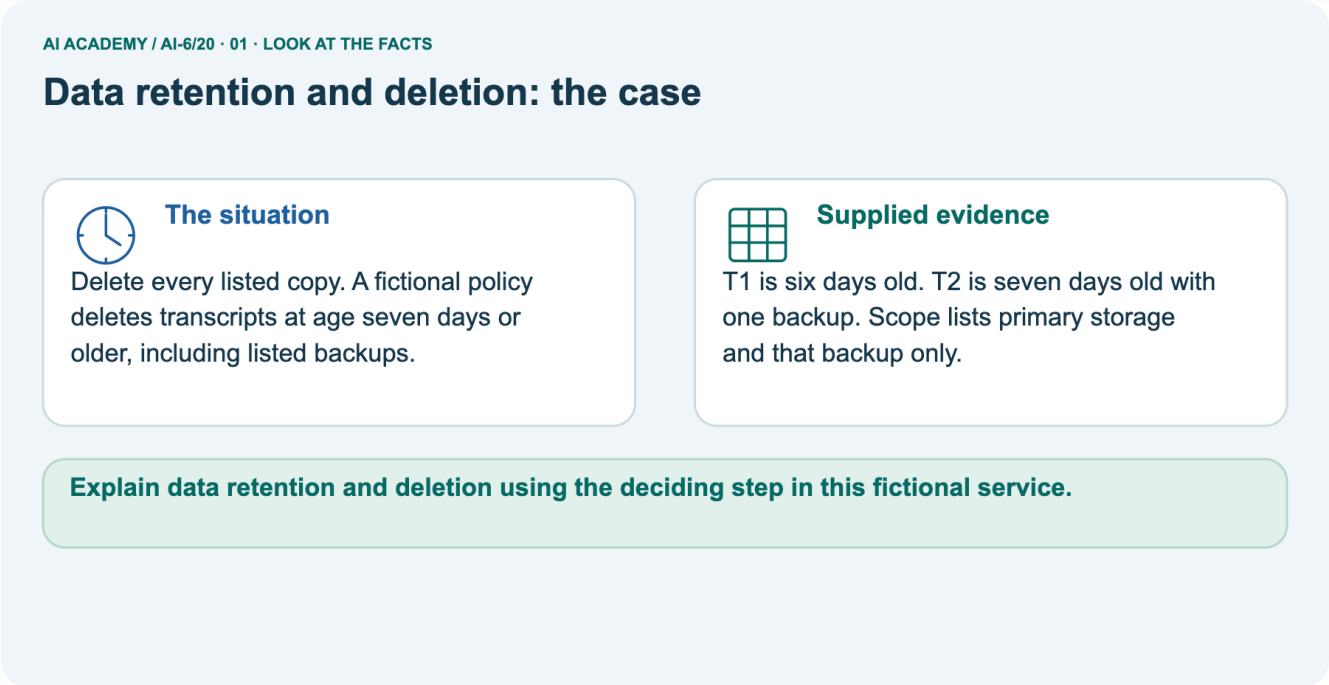
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

## Visual concept journey

> Explain each Data retention and deletion graphic using its labels, deciding evidence and relevant limitation.

### 01. Data retention and deletion: the case



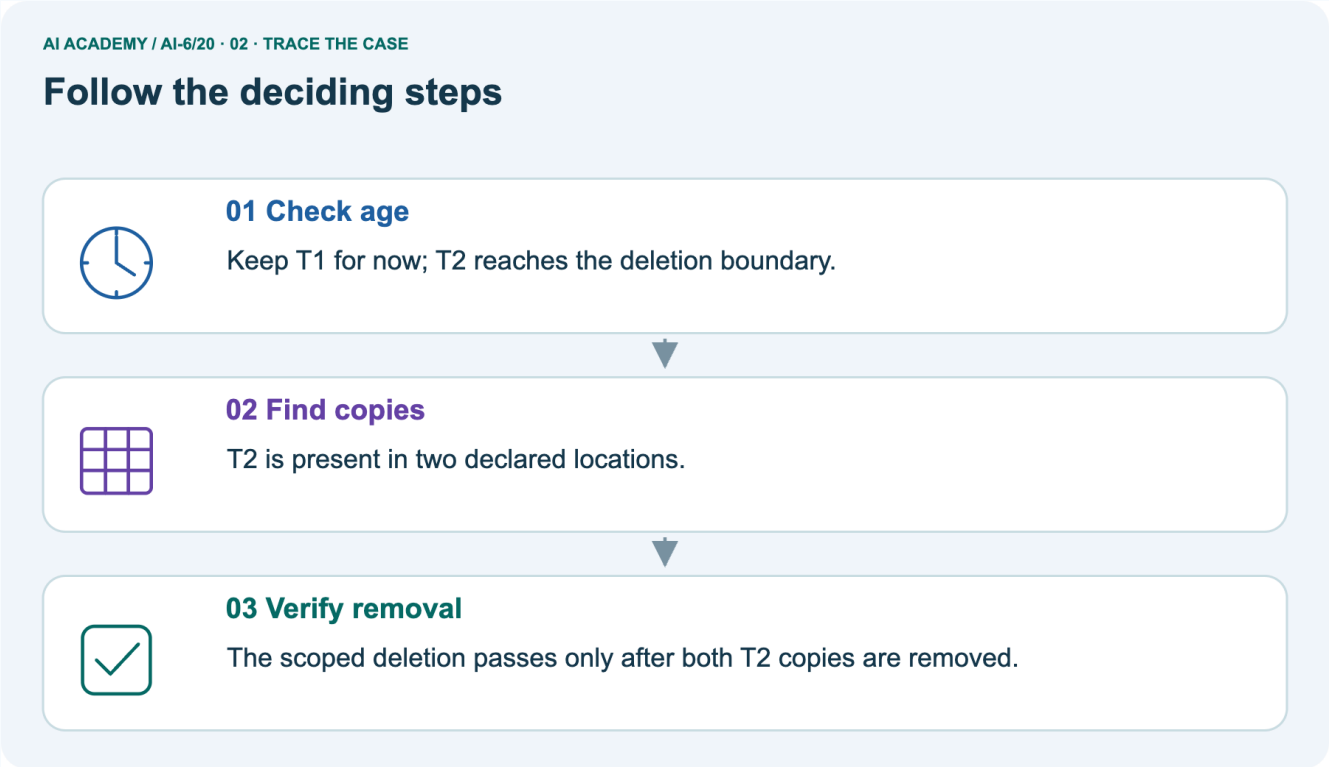

Follow the labelled picture with the matching explanation. Open the vector diagram

Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups. Supplied facts: T1 is six days old. T2 is seven days old with one backup.

Scope lists primary storage and that backup only.

> Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps

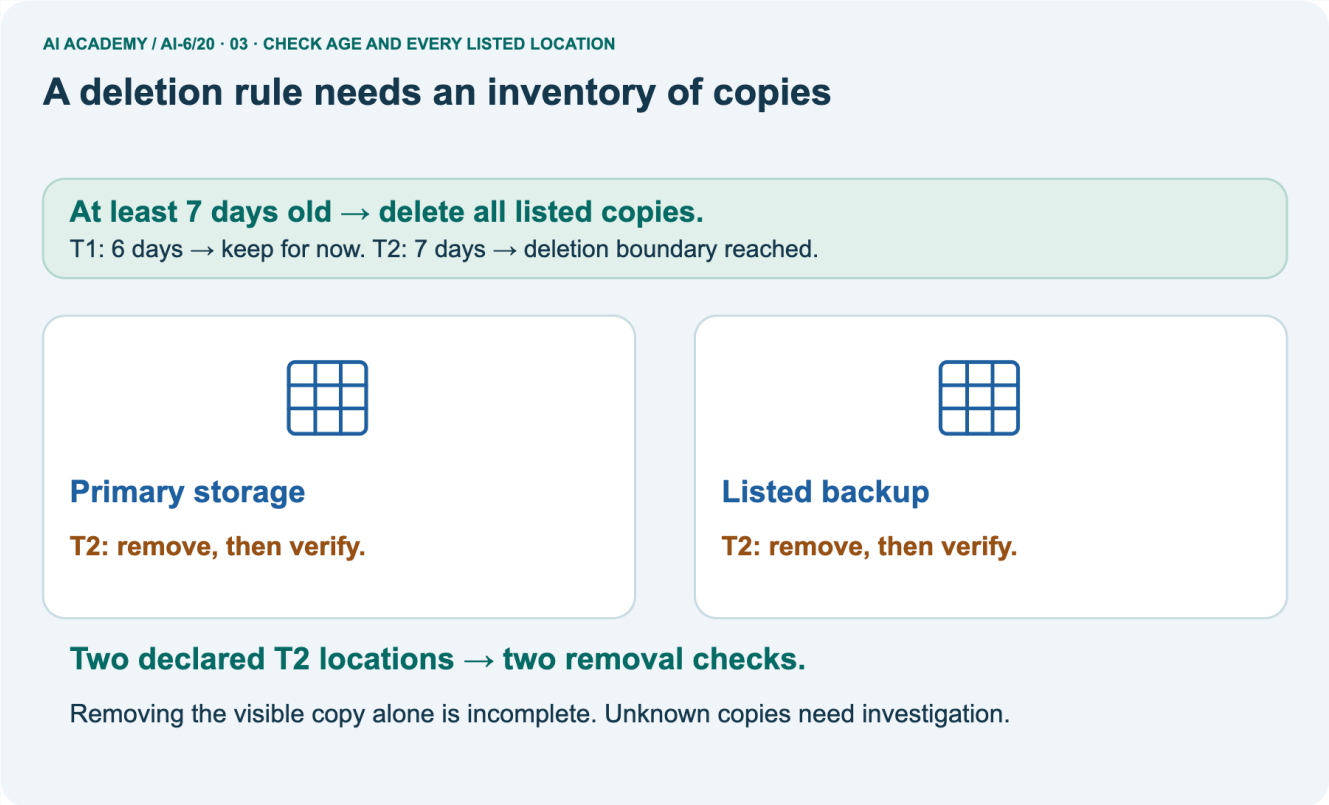



Follow the labelled picture with the matching explanation. Open the vector diagram

Check age: Keep T1 for now; T2 reaches the **deletion boundary**. → Find copies: T2 is present in two declared locations. → Verify removal: The scoped deletion passes only after both T2 copies are removed.

> Point to each numbered stage and explain which recorded **input** it uses.

## 03. A deletion rule needs an inventory of copies



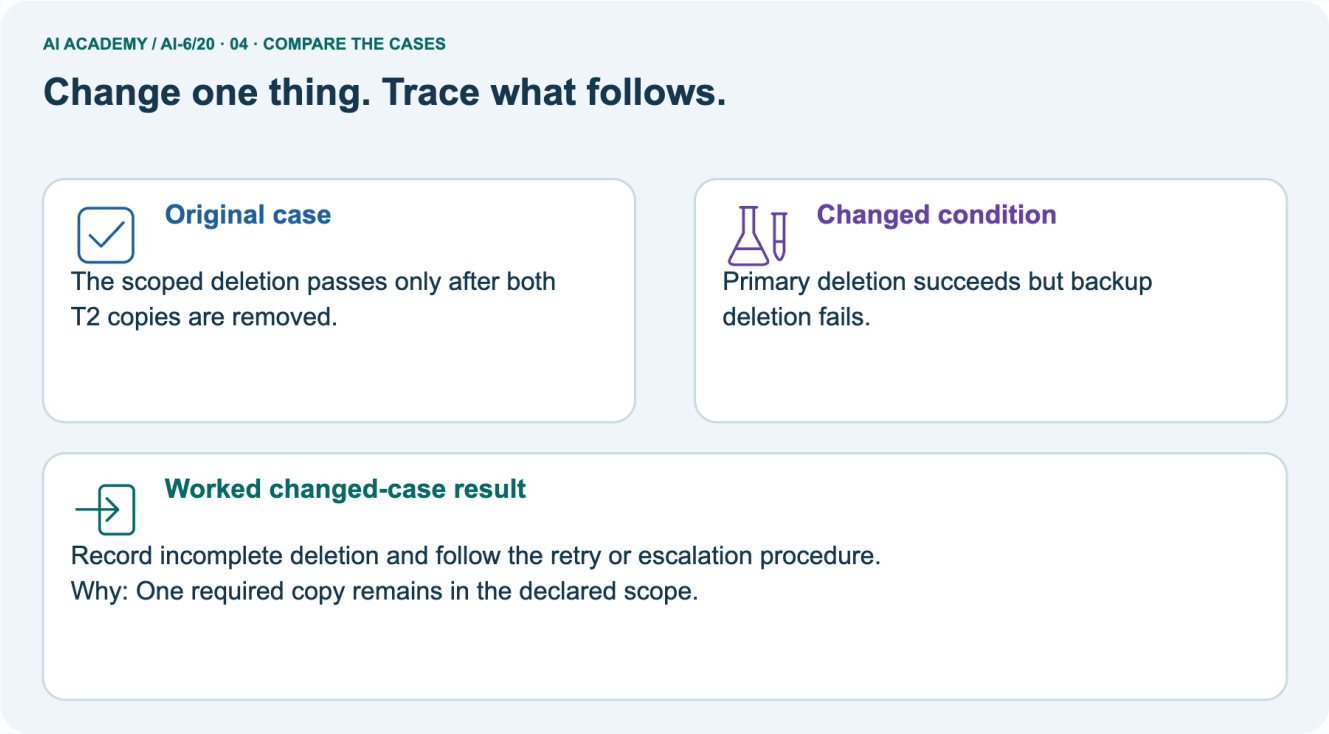

Follow the labelled picture with the matching explanation. Open the vector diagram

The toy retention boundary is seven days or older. Keep six-day T1 for now. Seven-day T2 appears in primary storage and one listed backup; both locations need removal verification.

This diagram specifies required checks, not an actual deletion run.

> Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



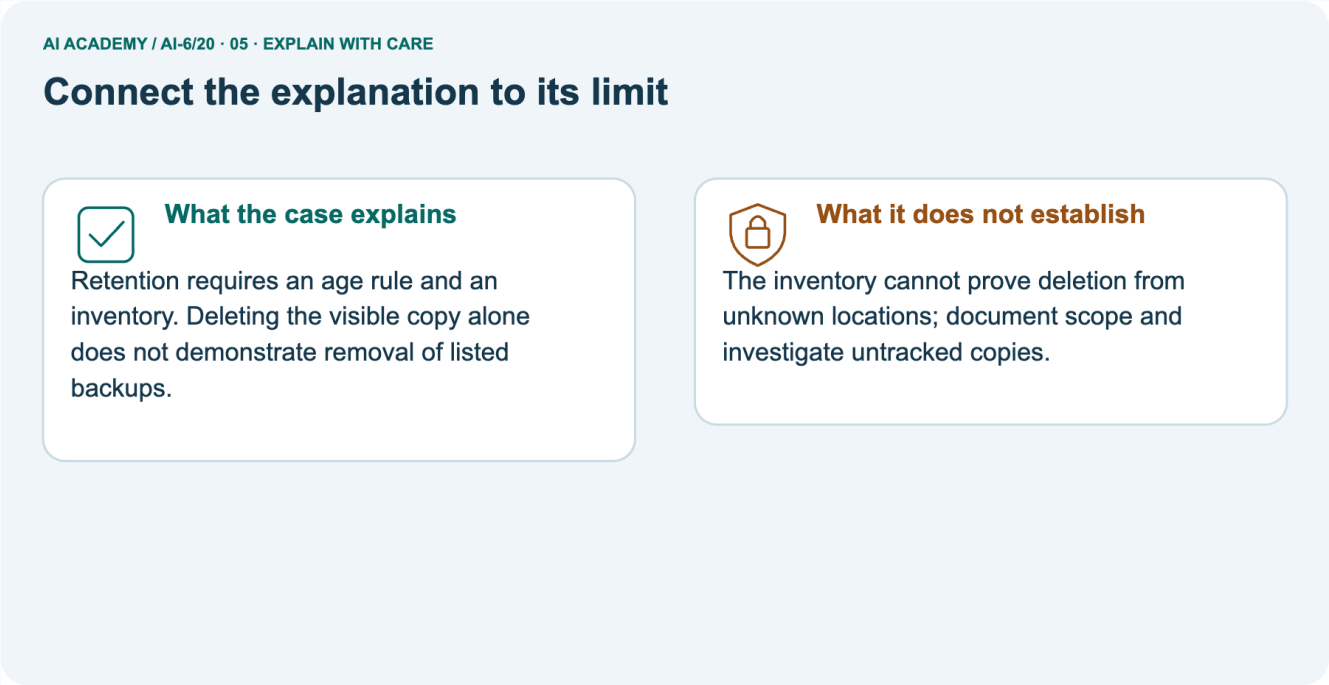

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The scoped deletion passes only after both T2 copies are removed. Changed condition: Primary deletion succeeds but backup deletion fails.

Expected result: Record incomplete deletion and follow the retry or escalation procedure. Reason: One required copy remains in the declared scope.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit



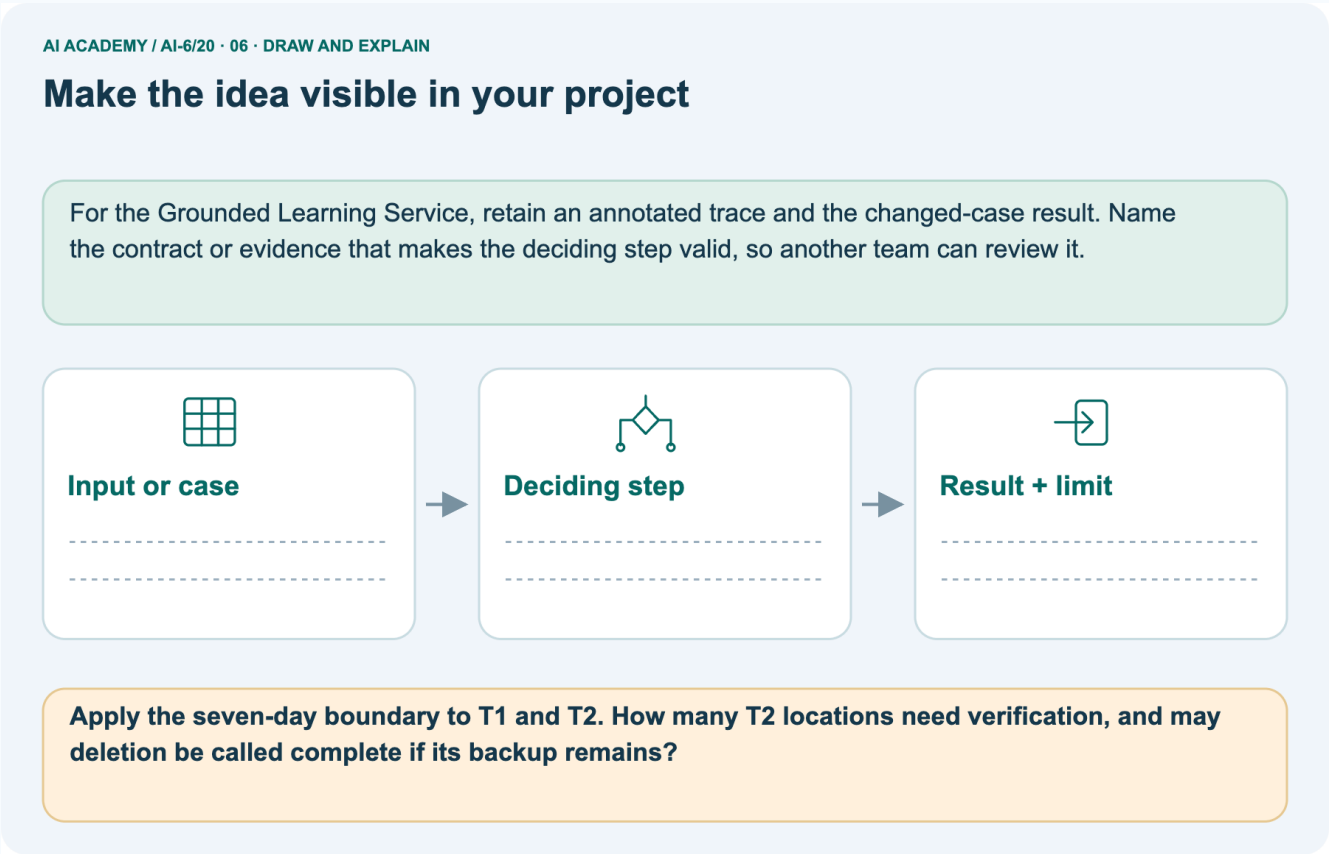

Follow the labelled picture with the matching explanation. Open the vector diagram

Retention requires an age rule and an inventory. Deleting the visible copy alone does not demonstrate removal of listed backups.

Limit: The inventory cannot prove deletion from unknown locations; document scope and investigate untracked copies.

> Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the seven-day boundary to T1 and T2. How many T2 locations need verification, and may deletion be called complete if its backup remains?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.